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L19: Index Arithmetic and n th power $\textcircled{1}$ residues

- We will give an application of primitive roots to solving certain congruences.
- Let $m \in \mathbb{Z}$ with $m > 0$ such that m has a primitive root, say r . Then $\{r, r^2, \dots, r^{\phi(m)}\}$ is a complete reduced residue system modulo m .

Definition: Let r be a primitive root modulo m . If $a \in \mathbb{Z}$ with $(a, m) = 1$, then the index of a relative to r , denoted $\text{ind}_r a$, is the least positive integer n for which $r^n \equiv a \pmod{m}$.

- Note that $\text{ind}_r a$ exists and $1 \leq \text{ind}_r a \leq \phi(m)$, by the observation made before the definition.

Example: 3 is a primitive root modulo 7 and

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$$3^1 \equiv 3 \pmod{7} \Rightarrow \text{ind}_3 3 = 1;$$

$$3^2 \equiv 2 \pmod{7} \Rightarrow \text{ind}_3 2 = 2;$$

$$3^3 \equiv 6 \pmod{7} \Rightarrow \text{ind}_3 6 = 3;$$

$$3^4 \equiv 4 \pmod{7} \Rightarrow \text{ind}_3 4 = 4;$$

$$3^5 \equiv 5 \pmod{7} \Rightarrow \text{ind}_3 5 = 5;$$

$$3^6 \equiv 1 \pmod{7} \Rightarrow \text{ind}_3 1 = 6.$$

- Also note that if a and b are relatively prime to m and $a \equiv b \pmod{m}$, then $\text{ind}_r a = \text{ind}_r b$.
- Indices enjoy a lot of properties in common with logarithms (the primitive root plays the role of the base of the logarithm).

Proposition 5.20: Let r be a primitive root modulo m and let $a, b \in \mathbb{Z}$ with $(a, m) = 1$ and $(b, m) = 1$. Then

$$(a) \text{ind}_r 1 \equiv 0 \pmod{\phi(m)}$$

$$(b) \text{ind}_r r \equiv 1 \pmod{\phi(m)}$$

$$(c) \text{ind}_r(ab) \equiv \text{ind}_r a + \text{ind}_r b \pmod{\phi(m)} \quad (3)$$

(d) $\text{ind}_r(a^n) \equiv n \text{ind}_r a \pmod{\phi(m)}$, if n is a positive integer.

Proof: (a) and (b) are clear, and left as an exercise. For (c), by definition, we have $r^{\text{ind}_r a} \equiv a \pmod{m}$ and $r^{\text{ind}_r b} \equiv b \pmod{m}$.

Multiplying these congruences gives

$$r^{\text{ind}_r a + \text{ind}_r b} \equiv ab \pmod{m}$$

Since $r^{\text{ind}_r(ab)} \equiv ab \pmod{m}$, we have that $r^{\text{ind}_r(a) + \text{ind}_r(b)} \equiv r^{\text{ind}_r(ab)} \pmod{m}$.

Proposition 5.2 gives

$$\text{ind}_r a + \text{ind}_r b \equiv \text{ind}_r(ab) \pmod{\phi(m)},$$

as desired. For (d), by definition we have

$$r^{\text{ind}_r(a^n)} \equiv a^n \pmod{m}. \text{ Also we have that}$$

$$r^{n \text{ind}_r a} \equiv (r^{\text{ind}_r a})^n \equiv a^n \pmod{m}.$$

Thus $r \operatorname{ind}_r(a^n) \equiv r^n \operatorname{ind}_r a \pmod{m}$, (4)
and Proposition 5.2 gives

$$\operatorname{ind}_r(a^n) \equiv n \operatorname{ind}_r a \pmod{\phi(m)},$$

as desired. □

Example: We illustrate Proposition 5.20(c) with an example. Modulo 7, we have that $\operatorname{ind}_3 2 = 2$ and $\operatorname{ind}_3 3 = 1$. By Proposition 5.20(c), we have

$$\begin{aligned} \operatorname{ind}_3 6 &= \operatorname{ind}_3(2 \cdot 3) = \operatorname{ind}_3 2 + \operatorname{ind}_3 3 \pmod{\phi(7)} \\ &\equiv 2 + 1 \pmod{6} \\ &\equiv 3 \pmod{6}. \end{aligned}$$

So $\operatorname{ind}_3 6 = 3$, which agrees with our previous computation!

- Suppose that $m \in \mathbb{Z}$ with $m > 0$ such that m has a primitive root, say r . Let $a, b \in \mathbb{Z}$ with $(a, m) = (b, m) = 1$. For $n \in \mathbb{Z}$ with $n > 0$, consider the congruence $ax^n \equiv b \pmod{m}$ (1).

We have $\text{ind}_r(ax^n) = \text{ind}_r(b)$. Since ⑤

$$\begin{aligned}\text{ind}_r(ax^n) &\equiv \text{ind}_r a + \text{ind}_r(x^n) \pmod{\phi(m)} \\ &\equiv \text{ind}_r a + n \text{ind}_r x \pmod{\phi(m)}\end{aligned}$$

by Proposition 5.20 (b) and (c) respectively, we obtain,

$$\text{ind}_r a + n \text{ind}_r x \equiv \text{ind}_r b \pmod{\phi(m)}.$$

This is equivalent to

$$n \text{ind}_r x \equiv \text{ind}_r b - \text{ind}_r a \pmod{\phi(m)}. \quad \textcircled{2}$$

Solving ① is equivalent to solving ②. However, ② is a linear congruence in the variable $\text{ind}_r x$ and can be solved using the techniques learned at the start of the course!

Example: Use indices to find all incongruent solutions to $6x^4 \equiv 3 \pmod{7}$.

We previously showed that 3 is a primitive root modulo 7. This is equivalent to the

$$\text{congruence } 4 \text{ind}_3 x \equiv \text{ind}_3 3 - \text{ind}_3 6 \pmod{\phi(7)}. \quad \textcircled{3}$$

Since $\text{ind}_3 3 = 1$, $\text{ind}_3 6 = 3$, and $\phi(7) = 6$, (6)
the congruence (3) becomes

$$4 \text{ind}_3 x \equiv 4 \pmod{6}.$$

Thus $2 \text{ind}_3 x \equiv 2 \pmod{3},$

and $\text{ind}_3 x \equiv 1 \pmod{3}.$

Thus $\text{ind}_3 x \equiv 1 \pmod{6}$

or $\text{ind}_3 x \equiv 4 \pmod{6}.$

Thus $x \equiv 3^1 \equiv 3 \pmod{7}$

or $x \equiv 3^4 \equiv 4 \pmod{7}.$

There are two incongruent solutions to the congruence $6x^4 \equiv 3 \pmod{7}.$

Definition: Let $a, m, n \in \mathbb{Z}$ with $m, n > 0$ and $(a, m) = 1$. Then a is said to be an n th power residue modulo m if the congruence $x^n \equiv a \pmod{m}$ is solvable in $\mathbb{Z}$.

Example: • 6 is a third power residue modulo 7 since the congruence $x^3 \equiv 6 \pmod{7}$ is solvable ($x=3$ is a solution). (7)

• 3 is a 4th power residue modulo 13 since the congruence $x^4 \equiv 3 \pmod{13}$ is solvable ($x=2$ is a solution).

• 3 is not a 4th power residue modulo 7 since the congruence $x^4 \equiv 3 \pmod{7}$ has no solutions (check this for yourself).

Theorem 5.21: Let $a, m, n \in \mathbb{Z}$ with $m > 0$ and $(a, m) = 1$. If the modulus m has a primitive root, then a is an n th power residue modulo m if and only if

$$a^{\phi(m)/d} \equiv 1 \pmod{m}$$

where $d = (n, \phi(m))$. Furthermore, if a is an n th power residue modulo m , then the

Congruence $x^n \equiv a \pmod{m}$ has exactly d incongruent solutions modulo m .

Proof: Let r be a primitive root modulo m . (8)

Then the congruence

$$x^n \equiv a \pmod{m} \quad (4)$$

is equivalent to the congruence

$$n \operatorname{ind}_r x \equiv \operatorname{ind}_r a \pmod{\phi(m)}. \quad (5)$$

If $d = (n, \phi(m))$, then (5) is solvable in $\operatorname{ind}_r x$ if and only if $d \mid \operatorname{ind}_r a$ by Theorem 2.6. Observe that $d \mid \operatorname{ind}_r a$ if and only if

$$\left(\frac{\phi(m)}{d}\right) \operatorname{ind}_r a \equiv 0 \pmod{\phi(m)}$$

which is equivalent to the congruence

$$a^{\phi(m)/d} \equiv 1 \pmod{m}.$$

This proves the first part of the Theorem.

Furthermore, Theorem 2.6 implies that if $d \mid \operatorname{ind}_r a$, then (5) has exactly d solutions modulo $\phi(m)$.

Then (4) has exactly d solutions modulo m .

□

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Corollary 5.22: Let p be an odd prime number and let $a \in \mathbb{Z}$ be such that $p \nmid a$. Then a is a quadratic residue modulo p if and only if

$$a^{(p-1)/2} \equiv 1 \pmod{p}.$$

Furthermore, if a is a quadratic residue modulo p , the congruence $x^2 \equiv a \pmod{p}$ has exactly two incongruent solutions modulo p .

Proof: Let $m=p$ and $n=2$ in Theorem 5.21.

Example: Let $a=6$, $m=7$, and $n=3$ in Theorem 5.21. A primitive root exists mod 7 by Theorem 5.19. Then $d = (3, \phi(7)) = 3$. By Theorem 5.21 we have that 6 is a 3rd power residue modulo 7 if and only if $6^{6/3} \equiv 1 \pmod{7}$. Since this congruence is true, 6 is a 3rd power

residue modulo 7 i.e. $x^3 \equiv 6 \pmod{7}$ (10)
is solvable. It has exactly three
incongruent solutions $\pmod{7}$ by Theorem
5.21.

Exercise: Is 3 a 4th power residue modulo
13? If so, how many incongruent solutions
does $x^4 \equiv 3 \pmod{13}$ have?

Example: Find all 15th power residues
modulo 9. A primitive root modulo 9
exists by Theorem 5.19. Then $d = (15, \phi(9))$
 $= 3$, and by Theorem 5.21, we have that
 a is a 15th power residue modulo 9 if
and only if $a^{6/3} \equiv 1 \pmod{9}$, or equivalently
 $a^2 \equiv 1 \pmod{9}$. This has solutions $a \equiv 1, 8 \pmod{9}$.
Thus the two 15th power residues mod 9 are
1 and 8. Also note that the 15th
power residues mod 9 are precisely the

3rd power residues (mod 9)

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i.e. by Euler's Theorem we have that

$a^6 \equiv 1 \pmod{9}$ for all $a \in \mathbb{Z}$ with $(a, 9) = 1$,

and so $a^{15} \equiv a^3 \pmod{9}$.